

# Sheldon George Bruno Waugh, MSc, PhD

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## Relevant work experience

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**Lead Data Scientist** | U.S. Department of Commerce, Office of Inspector General (DOC OIG) | Washington DC  
Series: 1560 - Grade: 14  
06/2023 - Present  
Hours per week: 40

Serves as the chief and principal data scientist for the DOC OIG, leading the Data Analytics team in data science research and projects to drive data-informed decisions.

- Developed, implemented and sustained IT infrastructure modernization effort for DOC OIG, enacting a on-prem analytics platform utilizing Posit Workbench processing with the capacity to run and validate frontier models from NIST and querying TBs of geospatial, financial and miscellaneous data from numerous systems within the Department of Commerce.
- Modernized the data lifecycle for unstructured grant documentation by piloting Natural Language Processing (NLP) and Generative AI workflows. This project converted raw text into structured data and utilized Agent Text-to-SQL generation via LLMs, enabling investigators to detect pass-through fraud and discover novel insights in broadband deployment data.
- Developed, built and sustained R package *Gumshoe* that streamlines and automates data reliability through Exploratory Data analysis and developing comprehensive reports for audit team use.
- Developed, created, administered and maintained multiple ArcGIS Online dashboards to monitor audit and investigative key programmatic areas, including Infrastructure Investment and Jobs Act (IIJA) broadband deployment, NOAA hurricane errors, and Pandemic Response Accountability Committee (PRAC) Paycheck Protection Program (PPP) loan usage.
- Builds, manages, and automates complex data pipelines in an Agile project-oriented customer-focused environment and establishes a division-level ArcGIS Online server environment to streamline metric calculations, data uploads, and visualizations.
- Architected and promulgated the Data Etiquette Guidelines for DOC OIG, establishing the agency's data operating model. These policies set best practices for data lifecycle management, curation, and metadata standardization for server-based databases supporting oversight reports.
- Designed and maintained the agency's geospatial data catalog in ArcGIS Online. Automated the collection of 4 TBs of geospatial metadata through API scraping, creating a unified data asset registry to ensure compliance with the Geospatial Data Act.
- Led OIG's participation in the PRAC data sharing pilot, using an open analytics platform to integrate data assets and deploy analytics solutions at scale.

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**Data Scientist** | U.S. Census Bureau | Suitland, MD  
Series: 1560 - Grade: 13  
01/2021 - 06/2023  
Hours per week: 40

- Served as Senior Administrator for the Officer of Survey and Census Analytics (OSCA) Tableau Server, overseeing the division's data catalog. Implemented data governance by defining roles, managing permissions, and organizing content to maintain protected access for the Field Quality Monitoring Program.

- Led development of metadata and visualization guidelines for all Business Intelligence platforms. Collaborated with data engineering teams to design schemas and data dictionaries, providing uniform metric calculations across the Unified Tracking System (UTS) data warehouse.

- Coordinated inventory, extraction, and transformation of complex survey data (CARI/SIPP), managing the lifecycle from raw audio files to structured datasets used for anomaly detection and fraud prevention.

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**Epidemiologist** | U.S. Army Public Health Center | Edgewood, MD  
Series: 0601 - Grade: 12  
06/2018 - 01/2021  
Hours per week: 40

- Led a directorate-wide initiative to create a comprehensive data dictionary and glossary. This project defined structure, information flow, and access standards for all surveillance data assets, strengthening the organization's data governance.

- Served as Principal Investigator for a \$350K grant, spearheading the integration of high-performance computing, machine learning, and artificial intelligence into the Army Veterinary Services to modernize public health surveillance.

- Managed the data lifecycle for the Military Working Dog (MWD) repository. Consolidated EHR, laboratory, and corporate data into a single enterprise inventory, creating a unified epidemiologic database for the DoD.

- Served as Veterinary Liaison for the Digital Health Strategic Initiative, leading planning and budgeting to advance data governance and management across the Army Veterinary Service.

## Education, certification or licenses

University of Florida | Gainesville, FL  
Doctorate degree  
Major: Epidemiology

Completion date: 05/2018

University of Florida | Gainesville, FL  
Master's degree  
Major: Medical Geography

Completion date: 05/2014

University of Florida | Gainesville, FL  
Bachelor's degree  
Major: Geography | Minor: Military Science

Completion date: 12/2011

## Professional publications

Bayko, H.; Watkins, S.; **Waugh, S.**; Moore, G.; Mullaney, S. Adaptation of the One Health Zoonotic Disease Prioritization Tool for Government- and Privately Owned Companion Animal Zoonotic Disease Surveillance. *Zoonotic Dis.* 2023, 3, x. <https://doi.org/10.3390/zoonoticdis3030020>

Seal, L. M., Mullaney, S. B., & **Waugh, S. G.** (2022). Leishmaniasis in the United States military veterinary patient population. *Journal of the American Veterinary Medical Association*, 1(aop), 1-4.

Tagliamonte, M. S., **Waugh, S.**, Prosperi, M., Mai, V. (2019, September). An Integrated Approach for Efficient Multi-Omics Joint Analysis. In *Proceedings of the 10th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics* (pp. 619-625). ACM.

**Waugh, S.**, and Mullaney, S. "Progress towards Companion Animal Zoonotic Disease Surveillance in the US Army." *Online Journal of Public Health Informatics* 11.1 (2019).

Ball, J. D., Fe Agana, D., **Waugh, S.**, Wang, K., James, T. G., Nicolette, G. (2019). Systematically collected information at encounters with HIV-positive students: A review of 10 years of electronic medical records. *Journal of American College Health*, 1-5. PMID: 30681932

Jennifer C. Dennis, Tyler Culpepper, Carmelo Nieves, Jr., Cassie C. Rowe, Alyssa M. Burns, Carley T. Rusch, Ashton Federico, Maria Ukhanova, **Waugh, S.**, Volker Mai, Mary C. Christman, Bobbi Langkamp-Henken, Probiotics (Lactobacillus gasseri KS-13, Bifidobacterium bifidum G9-1, and Bifidobacterium longum MM-2) improve rhinoconjunctivitis-specific quality of life in individuals with seasonal allergies: a double-blind, placebo-controlled, randomized trial. *Am J Clin Nutr* 105, 758–767 (2017). PMID: 28228426

Oliveira, F.P. de, Mendes, R.H., Dobbler, P.T., Mai, V., Pylro, V.S., **Waugh, S.**, Vairo, F., Refosco, L.F., Roesch, L.F.W., and Schwartz, I.V.D. (2016). Phenylketonuria and Gut Microbiota: A Controlled Study Based on Next-Generation Sequencing. *PLOS ONE* 11, e0157513. PMID: 27336782

## Additional information

### Technical Skills:

- Data Governance: Data Dictionaries, Metadata Standards, Data Lifecycle Management, Data Glossaries. Data Cataloging: ArcGIS Online, Tableau Server, CKAN/DKAN concepts, Data Inventorying.
- Tools & Languages: Python (Pandas, Scikit-learn, ArcPy), R (Tidyverse), SQL, SAS, ESRI ArcGIS, Tableau, PowerBI.

### Military Experience:

- Major, Signal Corps, United States Army Reserve (2011–Present)
  - Deputy Chief of Staff, G6 (412<sup>th</sup> Theater Engineer Command): Manage enterprise communications networks and signal operations.
  - Brigade Communications Officer: Manage enterprise communications networks and signal operations.
  - Previous Command: Company Commander (2015-2018).

Security Clearance: TS/SCI

### Selected Awards:

University of Florida Outstanding Young Alumni Award (40 Gators under 40) (2025)

Department of Commerce Bronze Award (2023)

Director's Award for Innovation, U.S. Census Bureau (2022)

Army Medicine Annual Wolf Pack Award for FY21 (2021)

Army Commendation Medal (2022, 2019)

SMART Scholarship, Department of Defense (2016–2018)

McKnight Fellowship, Florida Education Fund (2014–2018)